Ordered Pairs

The **ordered pair** is written in the form $(a\,,b)$. a called the **first order** (**component**) and b the **second order** (**component**). if $(a,b)=(c,d)$ then $a=c\,;\,b=d$

Example

Find the value of x and y if $(2,x-y)=(x+y,6)$

Solution

$$x+y=2 \qquad (i)$$
$$x-y=6 \qquad (ii)$$

From (i) and (ii) , we have $x=4\,,\ \ y=-2$

Note

- A is called an **unordered pair**, if $A=\{x,y\}$ for some x,y if A has exactly **two elements**.
- A is called a **singleton** if $A=\{x\}$ for some x, i.e. if A has exactly **one element**.

Cartesian Product

Let A and B be sets. The set of **all orderd pairs** having **first coordinate** in A and **scond coordinate** in B is called **Cartesian product** of A and B, denoted by $\boldsymbol{A}\times\boldsymbol{B}$ and read A ***cross*** B

$$A\times B=\{(a,b):a\in A\,,b\in B\}$$

Example: If $A=\{x,y\}$ and $B=\{1,3,4\}$, then

$$A\times B=\{(x,1),(x,3),(x,4),(y,1),(y,3),(y,4)\}$$
$$B\times A=\{(1,x),(1,y),(3,x),(x,y),(4,x),(4,y)\}$$

Note: $A\times B\neq B\times A$

If $|A|=n\ ;\ |B|=m$ then $|A\times B|=|B\times A|=n.m$

Example: if $A=\{a,b\}$, find $A\times\mathcal{P}(A)$

Solution

$\mathcal{P}(A)=\{\phi,A,\{a\},\{b\}\}$

$A\times\mathcal{P}(A)=\{(a,\phi),(a,A),(a,\{a\}),(a,\{b\}),(b,\phi),(b,A),(b,\{a\}),(b,\{b\})\}$

Relations

Let A and B be sets. The **Relation** from A to B is a **subset of** $\boldsymbol{A \times B}$**,** denoted by $\boldsymbol{R}$

Example

If $A = \{x, y\}$; $B = \{1,3,4\}$ and $R = \{(x, 1), (x, 4), (y, 1), (y, 3)\}$

(1) $(x, 1) \in R \quad \rightarrow xR1 \qquad x{\sim}1$

Reading x has **a relation** with 1

(2) $(x, 3) \notin R \quad \rightarrow x \not R 3 \qquad x \nsim 3$

Reading x has **not a relation** with 3

Example

If $A = \{1,2,3\}$; $B = \{1,2,4, 5\}$, find the following

(1) $A \times B$

$A \times B = \{(1,1), (1,2), (1,4), (1,5), (2,1), (2,2), (2,4), (2,5), (3,1), (3,2), (3,4), (3,5)\}$

(2) $R_1 = \{(x, y): (x, y) \in A \times B \quad \land \quad x = y\}$

$R_1 = \{(1,1), (2,2)\}$

(3) $R_2 = \{(x, y): (x, y) \in A \times B \quad \land \quad x < y\}$

$R_2 = \{(1,2), (1,4), (1,5), (2,4), (2,5), (3,4), (3,5)\}$

(4) $R_3 = \{(x, y): (x, y) \in A \times B \quad \land \quad x > y\}$

$R_3 = \{(2,1), (3,1), (3,2)\}$

(5) $R_4 = \{(x, y): (x, y) \in A \times B \quad \land \quad x = y + 1\}$

$R_4 = \{(2,1), (3,2)\}$

(6) $R_5 = \{(x, y): (x, y) \in A \times B \quad \land \quad x \backslash y\}$; x divides y

$R_5 = \{(1,1), (1,2), (1,4), (1,5), (2,2), (2,4)\}$

Invers Relation

If R a relation from A to B , the **inverse relation** denoted by R^{-1} and define as

$$R^{-1} = \{(y, x): (x, y) \in R\}$$

Example

If $A = \{1,2,3\}$; $B = \{1, 3\}$, find the following

(1) $A \times B$

$A \times B = \{(1,1), (1,3), (2,1), (2,3), (3,1), (3,3)\}$

(2) $R = \{(a, b): a > b\}$

$R = \{(2,1), (3,1)\}$

(3) R^{-1}

$R^{-1} = \{(1,2), (1,3)\}$

Definition

(i) The **domain** of the relation $\boldsymbol{R}$ from A to B is the set

Dom($\boldsymbol{R}$) = {x ∈ A: There exists some y ∈ B such that (x, y) ∈ $\boldsymbol{R}$}

(ii) The **range** of the relation $\boldsymbol{R}$ is the set

Rang($\boldsymbol{R}$) = {y ∈ B: There exists some x ∈ A such that (x, y) ∈ $\boldsymbol{R}$}

(iii) The **field** of the relation $\boldsymbol{R}$ is

Fld($\boldsymbol{R}$) = Dom($\boldsymbol{R}$) ∪ rang($\boldsymbol{R}$)

Example

Let A={1, 2, 3, 4, 5, 6}, and define R by xRy iff $x < y$ and x divides y; find the **domain, range** and **field** of the relation R

Solution

$R = \{(1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 4), (2, 6), (3, 6)\}$

$\text{Dom}(R) = \{1, 2, 3\}$

$\text{Rang}(R) = \{2, 3, 4, 5, 6\}$

$\text{Fld}(R) = \text{Dom}(R) \cup \text{rang}(R) = \{1, 2, 3, 4, 5, 6\}$

Example

If $A = \{1,2,4\}$; $B = \{2, 3, 5\}$, and $R = \{(1,2), (1,3), (2,2)\}$; prove that

(i) $\text{Dom}(R^{-1}) = \text{Rang}(R)$

(ii) $\text{Rang}(R^{-1}) = \text{Dom}(R)$

(iii) $(R^{-1})^{-1} = R$

Proof

$R^{-1} = \{(2,1), (3,1), (2,2)\}$

$\text{Dom}(R) = \{1, 2\}$; $\text{Rang}(R) = \{2, 3\}$

$\text{Dom}(R^{-1}) = \{2, 3\}$; $\text{Rang}(R^{-1}) = \{1, 2\}$

Therefore, we get

(i) $\text{Dom}(R^{-1}) = \text{Rang}(R) = \{2, 3\}$

(ii) $\text{Rang}(R^{-1}) = \text{Dom}(R) = \{1, 2\}$

(iii) $(R^{-1})^{-1} = \{(1,2), (1,3), (2,2)\} = R$

Propertis of Relation

(1) Reflexive Relation

The relation R is a **Reflexive Relation** on a set A if

$$\forall x \in A \quad \rightarrow \quad (x, x) \in R$$

Example

- If $A = \{1, 2\}$ and $R = \{(1,1), (1,2), (2,2)\}$

  $\therefore$ R is reflexive

- If $B = \{x, y, z\}$ and $R = \{(x, x), (x, y), (y, y), (y, z), (z, x)\}$

  $\therefore$ R is **not reflexive**

Because; $z \in B \quad \rightarrow \quad (z, z) \notin R$

Examples of reflexive relations

(i) Relation of **equality** on any group.
(ii) Relation **divides** by any set of numbers.
(iii) **Parallelism** relation on any set of straight lines in one plane.

(2) Symmetric Relation

The relation R is a **Symmetric Relation** on a set A if

$$\forall (x, y) \in R \quad \rightarrow \quad (y, x) \in R$$

Example

- If $A = \{1, 2, 3\}$ and $R = \{(1,1), (1,2), (2,1), (3,2), (3,3), (2,3)\}$

  $\therefore$ R is **symmetric**

- If $A = \{1, a, b\}$ and $R = \{(1,1), (1, a), (a, a), (a, b), (b, a)\}$

  $\therefore$ R is **not symmetric**

  Because; $(1, a) \in R \quad \rightarrow \quad (a, 1) \notin R$

Examples of symmetric relations

(i) Relation of **equality** in any group.
(ii) **Orthogonal** relation in a group of straight lines in a plane.
(iii) Siblings relationship (**Brothers**).
(iv) Some names and letters (**Nun - Mim - Tut**).

(3) Non-Symmetric Relation

The relation R is **non-Symmetric Relation** on a set A if

$$\forall (x, y) \in R \quad \rightarrow \quad (y, x) \notin R$$

Example

- If $A = \{1, 2, 3, 4, 5\}$ and $R = \{(1,2), (3,2), (4,5)\}$

  $\therefore$ R is **non-symmetric**

- If $A = \{1, 2, 3\}$ and $R = \{(1,2), (3,2), (2,1)\}$

  $\therefore$ R is **not non-symmetric**

  Because; $(1, 2) \in R \quad \rightarrow \quad (2,1) \in R$

  $\therefore$ R is **not symmetric**

  Because; $(3, 2) \in R \quad \rightarrow \quad (2,3) \notin R$

Examples of non symmetric relation

(i) The relation (**greater than**) in a set of numbers.
(ii) The relation (**less than**) in the set of numbers.
(iii) The relation (**divide**) in a set of numbers.

(4) Anti symmetric Relation

The relation R is **Anti symmetric Relation** on a set A if

$$\forall (x, y) \in R \quad and \quad (y, x) \in R \quad \rightarrow x = y$$

Example

If $A = \{a, b, c\}$ and

$R_1 = \{(a, b), (a, c), (b, c)\}$; $R_2 = \{(b, c)\}$; $R_3 = \{(a, c), (c, a), (b, c)\}$

$\therefore$ R_1, R_2 are **Anti symmetric relation**

$\therefore$ R_3 is **not Anti symmetric relation**

Because; $(a, c) \in R_3$ and $(c, a) \in R_3$ $\quad a \neq c$

Examples of Anti symmetric relation

(i) The relation (**greater than** or **equal** to) in a set of numbers.
(ii) The relation (**less than** or **equal** to) in the set of numbers.
(iii) The relation (**divide**) in a set of numbers.
(iv) The **inclusion** relation in groups.

(5) Transitive Relation

The relation R is a **Transitive Relation** on a set A if

$$\forall (x, y) \in R \quad and \quad (y, z) \in R \quad \rightarrow (x, z) \in R$$

Example

- If $A = \{1,2\}$ and $R = \{(1,1), (1,2), (2,1), (2,2)\}$

  $\therefore$ R is **Transitive relation**

- If $A = \{1,2,3\}$ and $R = \{(1,2), (2,3)\}$

  $\therefore$ R is **not Transitive relation**

Because; $(1,2) \in R$ and $(2,3) \in R$ $\quad \rightarrow (1,3) \notin R$

Examples of non symmetric relation

(i) The relation (**greater than**) in a set of numbers.
(ii) The relation (**less than**) in the set of numbers.
(v) The relation (**divide**) in a set of numbers.
(vi) The **inclusion** relation in groups

(6) Equivalence Relation

The relation $\boldsymbol{R}$ is an **Equivalence Relation** on a set A iff $\boldsymbol{R}$ is

(i) Reflexive (ii) Symmetric (iii) Transitive

Example:

If $A = \{1,2,3\}$, show which of the following relations are equivalent

$R_1 = \{(1,1),(2,2),(1,3),(3,2),(3,1),(3,3)\}$

$R_2 = \{(1,1),(2,3),(1,3),(3,2),(3,1),(3,3)\}$

$R_3 = A \times A$

Solution

R_1 is **not equivalent** because it is **not symetric**

R_2 is **not equivalent** because it is **not reflexive**

R_3 is **equivalent**

(7) Partial Order Relation

The relation $\boldsymbol{R}$ is a **Partial Order Relation** on a set A iff $\boldsymbol{R}$ is

(i) Reflexive (ii) Anti-symmetric (iii) Transitive

Example:

If $A = \{1, 2, 3\}$, show whether the relation $(x, y) \in R$, if, $x \geq y$ defined on the set A is a **partial order relation**.

Solution

$R = \{(2,1),(3,1),(3,2),(1,1),(2,2),(3,3)\}$

(i) Reflexive Relation:

The relation is reflexive as $\forall\, a \in A, (a, a) \in R$, i.e. $(1,1),(2,2),(3,3) \in R$.

(ii) Anti-symmetric Relation:

The relation is antisymmetric as $\forall\ (a, b)$ and $(b, a) \in R$, we have $a = b$.

(iii) Transitive Relation:

The relation is transitive as $\forall\ (a, b)$ and $(b, c) \in R$, we have $(a, c) \in R$.

For example: $(3,2) \in R$ and $(2,1) \in R$, implies $(3,1) \in R$.

Therefor the relation R is a **partial order relation.**

Example

(a) The relation ⊆ of a set of inclusion is a **partial ordering** or any collection of sets since set inclusion has three desired properties:

1. A ⊆ A for any set A.
2. If A ⊆ B and B ⊆ A then B = A.
3. If A ⊆ B and B ⊆ C then A ⊆ C

(b) The relation ≤ on the set R of real number that is **Reflexive**, **Antisymmetric** and **transitive**.

Therefor the relation ≤ is a **Partial Order Relation.**

Functions

A function is an **ordered triple** $(\boldsymbol{f}, \boldsymbol{A}, \boldsymbol{B})$ such that

(1) A and B are sets, and $f \subseteq A \times B$,

(2) For every x ∈ A there is some y ∈ B such that $(x, y) \in f$

Functions are **usually** denoted by the letters f, g, h ….. and are written $f: A \to B$, where we call set A the **domain** and set B the **codomain**.

- If $(x, y) \in f$, we usually write $y = f(x)$, and call y the **image** of x under f.
- The set $\{y \in B$: There is an $x \in A$ such that $y = f(x)\}$ is called the **range** of f.
- If $A = B$ and $f(x) = x$ for all $x \in A$, the f is called the **identity function** on A, and it is denoted by id_A
- If $f(x) = b$ for all $x \in A$, then f is called a **constant function.**

Example

If $A = \{0, 1, 2, 3\}$ and $B = \{0, 2, 3, 4, 5\}$ $f: A \to B$, where $f(x) = x + 2$ find the **range** of function

Solution

$f(0) = 2$, $f(1) = 3$, $f(2) = 4$, $f(3) = 5$

∴ Rang$(f) = \{2, 3, 4, 5\}$

Example

If $f(x) = \frac{x+1}{x-1}$ find the image of following

(1) $f(2) = \frac{(2)+1}{(2)-1} = 3$

(2) $f(x-1) = \frac{(x-1)+1}{(x-1)-1} = \frac{x}{x-2}$

(3) $f\left(\frac{1}{x}\right) = \frac{\left(\frac{1}{x}\right)+1}{\left(\frac{1}{x}\right)-1} = \frac{\frac{1+x}{x}}{\frac{1-x}{x}} = \frac{1+x}{1-x}$

One - one, Onto, and Bijective Functions

(1) One-to-one (subjective) Function

The function $f: A \to B$ is said to be **a one-to-one** function if

$$f(x) = f(y) \leftrightarrow x = y$$

Example

Prove that the function $f: \mathbb{R} \to \mathbb{R}$, where $f(x) = 3x + 1$ is **One-to-one** function.

Proof

$f(x) = f(y) \leftrightarrow x = y$

$f(x) = 3x + 1 \quad ; \quad f(y) = 3y + 1$

$3x + 1 = 3y + 1$

$3x = 3y \quad \to \quad x = y \quad$ #

$\therefore$ f is **One-to-one** function

Example

Prove that the function $f: \mathbb{Z} \to \mathbb{Z}$, where $f(x) = x^2$ is **not One-to-one** function.

Proof

$f(x) = f(y) \leftrightarrow x = y$

$f(x) = x^2 \quad ; \quad f(y) = y^2$

$x^2 = y^2$

$x = \pm y \quad \to \quad x \neq y \quad$ #

$\therefore$ f is **not One-to-one** function

(2) Onto (injective) Function

The function $f: A \to B$ is said to be a **Onto function**, if for every $x' \in B$ there is $x \in A$ such that $f(x) = x'$

Example

Prove that the function $f: \mathbb{R} \to \mathbb{R}$, where $f(x) = 2x - 1$ is a onto function.

Proof

$f(x) = x'$

$2x - 1 = x' \qquad x = \frac{x'+1}{2}$

$\therefore$ $f(x) = 2\left(\frac{x'+1}{2}\right) - 1 = x' \qquad$ #

$\therefore$ f is **onto** function

(3) Bijective Function

The function $f: A \to B$ is said to be a **Bijective function** if and only if it is **one-to-one** and **onto** functions.

Composition Functions

Let $f: A \to B$ and $g: C \to D$ be functions such that $\text{rang}(f) \subseteq \text{dom}(g)$. Then the function $g \circ f: A \to C$ defined by $(g \circ f)(x) = g(f(x))$ is called the (functional) **composition** of f and g

Example

If $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ where $f(x) = 2x - 1$ and $g(x) = 3x + 1$, find the following

(i) $f \circ g$ (ii) $g \circ f$ (iii) $g \circ f \circ g$

Solution

(i) $$(f \circ g)(x) = f\big(g(x)\big) = f(3x + 1) = 2(3x + 1) - 1 = 6x + 1$$

(ii) $$(g \circ f)(x) = g\big(f(x)\big) = g(2x - 1) = 3(2x - 1) + 1 = 6x - 2$$

Note that $f \circ g \neq g \circ f$

(iii) $$(g \circ f \circ g)(x) = g[f \circ g] = g\big[f\big(g(x)\big)\big] = g[6x + 1] = 3(6x + 1) + 1 = 18x + 4$$

Inverse of Function

If $f: A \to B$, we define the **inverse of the function** with the relation $f^{-1}: B \to A$, provided that

$$f \circ f^{-1} = f^{-1} \circ f = I$$

Definition: $f: A \to B$ has an inverse if and only if f is **injective**.

Example: If $f: \mathbb{R} \to \mathbb{R}$ where $f(x) = 2x + 7$

(i) Prove that f^{-1} exists, then find it.

(ii) Prove that $f \circ f^{-1} = f^{-1} \circ f$

Proof

First, we prove that the function is **one-to-one.**

$$f(x) = f(y) \leftrightarrow x = y$$
$$f(x) = 2x + 7 \quad ; \quad f(y) = 2y + 7$$
$$2x + 7 = 2y + 7 \qquad 2x = 2y \qquad \to \quad x = y \qquad \#$$

Second, we prove that the function is **onto.**

$$f(x) = x'$$
$$2x + 7 = x' \qquad x = \frac{x' - 7}{2}$$
$$\therefore \quad f(x) = 2\left(\frac{x' - 7}{2}\right) + 7 = x' \qquad \#$$

$\therefore$ The inverse is existing

To find the inverse, we solve the equation $y = 2x + 7$ by **replacing** x for y and vice versa.

$$x = 2y + 7 \qquad y = \frac{1}{2}x - \frac{7}{2} \equiv f^{-1}(x)$$

(ii) Prove that $f \circ f^{-1} = f^{-1} \circ f$

$*$ $$(f \circ f^{-1})(x) = f\big((f^{-1})\big) = f\left(\frac{1}{2}x - \frac{7}{2}\right)$$
$$= 2\left(\frac{1}{2}x - \frac{7}{2}\right) + 7 = x$$

$*$ $$(f^{-1} \circ f)(x) = f^{-1}\big((f)\big) = f^{-1}(2x + 7)$$
$$= \frac{1}{2}(2x + 7) - \frac{7}{2} = x$$

$\therefore \quad f \circ f^{-1} = f^{-1} \circ f = x$ #

Example

If $f(x) = x - 1$ and $g(x) = 2x + 1$, find $h(x)$ in the following

$(i) \quad f \circ h = g \qquad (ii) \quad h \circ f = g$

Solution

$(i) \quad f \circ h = g$

Using the **identity function**, we find that

$$f^{-1} \circ f \circ h = f^{-1} \circ g$$
$$h = f^{-1} \circ g$$
$$= f^{-1}\big(g(x)\big) = f^{-1}(2x + 1)$$
$$= (2x + 1) + 1 = 2x + 2$$ #

$(ii) \quad h \circ f = g$

Using the **identity function**, we find that

$$h \circ f \circ f^{-1} = g \circ f^{-1}$$
$$h = g \circ f^{-1}$$
$$= g\big(f^{-1}(x)\big) = g(x + 1)$$
$$= 2(x + 1) + 1 = 2x + 3$$ #

Exercise (4)

(1) If $A = \{1,2,3,4,5,6\}$, discuss the properties of the following relations

$R_1 = \{(1,2),(3,2),(2,2),(2,3)\}$ $\qquad$ $R_2 = \{(1,2),(2,3),(1,3)\}$

$R_3 = \{(1,1),(2,2),(2,3),(3,2),(3,3)\}$ $\qquad$ $R_4 = \{(1,2)\}$ $\qquad$ $R_5 = A \times A$

(2) If $A = \big\{1,\{1\},\{1,\{1\}\}\big\}$, find the following

(i) $A \times A$ $\qquad$ (ii) $A \cap \mathcal{P}(A)$

(3) Find $g(f(x))$ *and* $f(g(x))$ of the following functions

(i) $f(x) = x^2 - 3x$; $\qquad$ $g(x) = \sqrt{x+5}$

(ii) $f(x) = \sqrt{5 - x^2}$; $\qquad$ $g(x) = \sqrt{x-3}$

(iii) $f(x) = \frac{x}{3x+2}$; $\qquad$ $g(x) = \frac{2}{x}$

(4) If $\emptyset(x) = \frac{x-1}{3x+5}$; find the following

(i) $\emptyset(1-x)$

(ii) $\emptyset\left(\frac{1}{x-1}\right)$

(iii) $\frac{1}{\emptyset(x)} + \emptyset(-1)$

(5) If $f(x-1) = 2x^2 + 5x - 7$; find the following

(i) $f(x)$

(ii) $f(5)$

(iii) $f(-1)$